Udit Rawat

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SUMMARY

AI Engineer with 2 years of production experience shipping LLMs where they are hardest to run: a diabetes-guidance SLM running fully offline on Snapdragon hardware in partnership with Eli Lilly, alongside production RAG and multi-agent systems, with evaluation frameworks, model routing and cost optimization built in from day one. Springer-published on LLM adaptation. Available to start immediately.

TECHNICAL SKILLS

Languages	Python (async), TypeScript, SQL, Shell
AI & ML	PyTorch, HF Transformers, scikit-learn, pandas, NumPy, LangChain, LangGraph, DSPy, RAG (pgvector, Qdrant), LLM Fine-tuning (LoRA/QLoRA), RAGAS
Model Optimization	GGUF, llama.cpp, ONNX, INT8/NF4 Quantization, Knowledge Distillation
Cloud & Backend	GCP, Azure, AWS, Cloudflare Workers/Pages, FastAPI, Docker, Kubernetes
MLOps & Data	LLM Gateway & Model Routing (LiteLLM), Observability (Langfuse), Structured Outputs (Instructor), ClickHouse, PostgreSQL, CI/CD

EXPERIENCE

TapHealth *New Delhi, India* Feb 2025 – Jul 2026
AI Engineer *(Apr 2025 – Jul 2026)*, previously **AI Intern** — promoted in under 2 months

- Built and shipped an offline diabetes-guidance SLM running fully on-device on Snapdragon 680 hardware, in partnership with Eli Lilly: 3× smaller at 380 MB, under 890 ms latency at 25 tok/s, 98% accuracy on out-of-distribution refusal. Fine-tuned Qwen 0.6B/1.5B with LoRA, quantized to GGUF Q4_K_M, with an ONNX INT8 classifier for OOD detection.
- Built a stateful 5-agent coaching system with LangGraph and Mem0/Hindsight memory across 25 tools, served via FastAPI with hybrid retrieval over pgvector/Qdrant. Built an async OCR/PDF pipeline streamed over SSE WebSockets, cutting report processing from 40s to 15s.
- Cut LLM costs 54% by designing a centralized LLM gateway (LiteLLM) with model routing and vendor selection across OpenAI, Anthropic, and open-source models for 65+ agentic workflows spanning 5+ microservices, with structured outputs via Instructor and end-to-end observability via Langfuse.
- Replaced a monolithic rules-based nudge engine with a decoupled DSPy GenAI architecture serving **5K+ users**, splitting habit classification from content generation, and built a ClickHouse-backed analytics service aggregating glucose, HbA1c and meal data, with full CI/CD.
- Started as AI Intern: built ARIMA/SARIMA/LSTM forecasting models for patient engagement and generated synthetic clinical data using NHANES + Q-learning, with LIME-based explainability.

Project Intern, Perception & Intelligence Lab, IIT Kanpur Jul 2024 – Oct 2024

- Curated a **200K+** image dataset and optimized diffusion-based inpainting models for a research lab, reaching 11.72 dB PSNR / 0.41 SSIM.

PROJECTS

Tribunal *(Python, LangGraph, MCP)* | github.com/udit-rawat/tribunal
Adversarial multi-agent fact-checker: parallel Prosecutor/Defender agents argue opposite sides over just-in-time RAG evidence (MiniLM embeddings, LanceDB), with a deterministic Citation Verifier striking any quote not found verbatim before a Judge rules. On a 32-claim balanced golden set: **96% quote grounding** (string match, not model judgement) and **83% polarity accuracy**. Shipped as CLI, MCP server and streaming FastAPI UI, with a RAGAS eval harness.

Hierarchical Reasoner *(Python, PyTorch)* | github.com/udit-rawat/hierarchical-reasoner
From-scratch PyTorch reproduction of the Hierarchical Reasoning Model (arXiv:2506.21734): a slow GRU planner driving a fast executor with ACT adaptive halting. **100% maze-path accuracy** by epoch 20 and **66.7% compute reduction** from ACT cutting H-steps 3 to 1, across 8 controlled experiments — including a documented negative result on 9×9 Sudoku, diagnosed as an architectural gap rather than tuning. 49/49 tests green in CI.

PUBLICATIONS

Efficient Adaptation of Lightweight LLMs with T-Free Tokenization ICAAI 2025, Springer
Boosting Algorithms for Class Imbalance SSWC 2025

EDUCATION

B.Tech in AI & ML, VIPS-TC, GGSIPU, New Delhi 2021 – 2025 *GPA: 9.1/10*